

Integration

Integration is the reverse of differentiation. Given the derivative of a function, the process of finding the original function is the reverse of differentiation, f is said to be an antiderivative of f' . Given

$$f'(x) = 4, \text{ then } f(x) = 4x + c.$$

Example 1

Find the antiderivative of any constant of $f'(x) = 0$.

Solution:

The derivative of any constant function is zero. Therefore, the antiderivative is $f(x) = c$.

Example 2

Find the antiderivative of $f'(x) = 2x - 7$.

Solution:

The antiderivative is $f(x) = x^2 - 7x + c$

Definition

If $F(x)$ is a function of x such that $\frac{d}{dx}F(x) = f(x)$, then we define integral of $f(x)$ with respect to (w.r.t) x to be the function $F(x)$ and we write

$$\int f(x)dx = F(x)$$

Thus $\frac{d}{dx}[F(x)] = f(x) \Rightarrow \int f(x)dx = F(x)$

For example $\frac{d}{dx}[x^5] = 5x^4$, we have $\int 5x^4dx = x^5$.

Moreover, if c is any constant, then $\frac{d}{dx}[x^5 + c] = 5x^4$.

So, in general, $\int 5x^4dx = x^5 + c$.

NOTE: Different values of c will give different integrals and thus integral of a function is unique.

Thus in general if f is a continuous function, then

$$\int f(x)dx = F(x) + c$$

If $F'(x) = f(x)$, where c is the constant of integration.

Rules of Integration

Rule 1. Constant functions

$$\int k dx = kx + c, \text{ where } k \text{ is any constant.}$$

Example

Evaluate the following integrals

$$(i) \quad \int -7 dx$$

$$(ii) \quad \int \frac{1}{3} dx$$

$$(iii) \quad \int \sqrt{3} dx$$

$$(iv) \quad \int -\frac{10}{7} dx$$

$$(v) \quad \int 0 dx$$

$$(vi) \quad \int \frac{\sqrt{3}}{2} dx$$

Solution:

$$(i) \quad \int -7 dx = -7x + c$$

$$(ii) \quad \int \frac{1}{3} dx = \frac{1}{3}x + c$$

$$(iii) \quad \int \sqrt{3} dx = \sqrt{3}x + c$$

$$(iv) \quad \int -\frac{10}{7} dx = -\frac{10}{7}x + c$$

$$(v) \quad \int 0 dx = 0x + c = c$$

$$(vi) \quad \int \frac{\sqrt{3}}{2} dx = \frac{\sqrt{3}}{2}x + c$$

Rule 2: Power Rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c, \text{ provided that } n \neq -1.$$

Example

Evaluate the following integrals

$$(i) \quad \int x dx$$

$$(ii) \quad \int x^2 dx$$

$$(iii) \quad \int \sqrt{x} dx$$

$$(iv) \quad \int \frac{1}{x^4} dx$$

$$(v) \quad \int \sqrt[3]{x} dx$$

$$(vi) \quad \int \sqrt[7]{x} dx$$

$$(vii) \quad \int \frac{1}{\sqrt{x}} dx$$

Solution:

$$(i) \quad \int x dx = \frac{x^{1+1}}{1+1} + c = \frac{x^2}{2} + c$$

$$(ii) \quad \int x^2 dx = \frac{x^{2+1}}{2+1} + c = \frac{x^3}{3} + c$$

$$(iii) \quad \int \sqrt{x} dx = \int x^{1/2} dx = \frac{x^{1/2+1}}{1/2+1} + c = \frac{x^{3/2}}{3/2} + c = \frac{2}{3} x^{3/2} + c$$

$$(iv) \quad \int \frac{1}{x^4} dx = \int x^{-4} dx = \frac{x^{-4+1}}{-4+1} + c = \frac{-1}{3} x^{-3} + c$$

$$(v) \quad \int \sqrt[3]{x} dx = \int x^{1/3} dx = \frac{x^{1/3+1}}{1/3+1} + c = \frac{x^{4/3}}{4/3} + c = \frac{3}{4} x^{4/3} + c$$

$$(vi) \quad \int \sqrt[7]{x} dx = \int x^{1/7} dx = \frac{x^{1/7+1}}{1/7+1} + c = \frac{x^{8/7}}{8/7} + c = \frac{7}{8} x^{8/7} + c$$

$$(vii) \quad \int \frac{1}{\sqrt{x}} dx = \int \frac{1}{x^{1/2}} dx = \int x^{-1/2} dx = \frac{x^{-1/2+1}}{-1/2+1} + c = \frac{x^{1/2}}{1/2} + c = 2x^{1/2} + c$$

Rule 3: Constant Times a Function

$$\int k f(x) dx = k \int f(x) dx, \text{ where } k \text{ is any constant.}$$

Example

Evaluate the following integrals

$$(i) \quad \int 4x dx$$

$$(ii) \quad \int \frac{x^2}{5} dx$$

$$(iii) \quad \int 3\sqrt{x} dx$$

$$(iv) \quad \int \frac{5}{\sqrt{x}} dx$$

$$(v) \quad \int 20x^5 dx$$

$$(vi) \quad \int 5\sqrt[3]{x} dx$$

$$(vii) \quad \int \frac{10}{\sqrt[5]{x^4}} dx$$

Solution:

$$(i) \quad \int 4x dx = \frac{4x^2}{2} + c = 2x^2 + c$$

$$(ii) \quad \int \frac{x^2}{5} dx = \frac{x^3}{5 \times 3} + c = \frac{x^3}{15} + c$$

$$(iii) \quad \int 3\sqrt{x} dx = \frac{3x^{3/2}}{3/2} + c = \frac{2}{3} \times 3x^{3/2} + c = 2x^{3/2} + c$$

$$(iv) \quad \int \frac{5}{\sqrt{x}} dx = \int 5x^{-1/2} dx = \frac{5x^{1/2}}{1/2} + c = \frac{2}{1} \times 5x^{1/2} + c = 10x^{1/2} + c$$

$$(v) \quad \int 20x^5 dx = \frac{20}{6} x^6 + c = \frac{10}{3} x^6 + c$$

$$(vi) \quad \int 5\sqrt[3]{x} dx = \int 5x^{1/3} dx = \frac{5}{1/3+1} x^{1/3+1} + c = \frac{5}{4/3} x^{4/3} + c = \frac{3}{4} \times 5x^{4/3} + c = \frac{15}{4} x^{4/3} + c$$

$$(vii) \quad \int \frac{10}{\sqrt[5]{x^4}} dx = \int 10x^{-4/5} dx = \frac{10x^{1/5}}{1/5} + c = 50x^{1/5} + c$$

Rule 4: Addition or Difference

If $\int f(x)dx$ and $\int g(x)dx$ exist, then $\int [f(x) \pm g(x)]dx = \int f(x)dx \pm \int g(x)dx$

Example

Evaluate the following integrals.

$$(i) \quad \int (3x - 5) dx$$

$$(ii) \quad \int (x^2 + 5x - 4) dx$$

$$(iii) \quad \int \left(6x^5 - \frac{2}{x^4} - 7x - 5 \right) dx$$

$$(iv) \quad \int \left(\sqrt{x} + \frac{1}{\sqrt[3]{x}} - 9x + 13 \right) dx$$

$$(v) \quad \int \left(\frac{x^4 - 3x^3 + x}{x} \right) dx$$

Solution:

$$(i) \quad \int (3x - 5) dx = \int 3x dx - \int 5 dx = \frac{3}{2}x^2 + c_1 - (5x + c_2) = \frac{3}{2}x^2 - 5x + c$$

NOTE: Even though the two integrals technically results in separate constants of integration, these constants may be considered together as one.

$$(ii) \quad \int (x^2 + 5x - 4) dx = \frac{x^3}{3} + \frac{5x^2}{2} - 4x + c$$

$$(iii) \quad \int \left(6x^5 - \frac{2}{x^4} - 7x - 5 \right) dx = x^6 + \frac{2}{3x^3} - \frac{7}{2}x^2 - 5x + c$$

$$(iv) \quad \int \left(\sqrt{x} + \frac{1}{\sqrt[3]{x}} - 9x + 13 \right) dx = \frac{2}{3}x^{3/2} + \frac{3}{2}x^{2/3} - \frac{9}{2}x^2 + 13x + c$$

$$(v) \quad \int \left(\frac{x^4 - 3x^3 + x}{x} \right) dx = \int (x^3 - 3x^2 + 1) dx = \frac{x^4}{4} - x^3 + x + c$$

Rule 5: Power Rule Exception

$$\int \frac{1}{x} dx = \int x^{-1} dx = \ln x + c$$

Rule 6: Exponential Function

$$\int e^{ax} dx = \frac{e^{ax}}{a} + c, \text{ where } a \text{ is a constant.}$$

Example

Evaluate the following integrals.

$$(i) \quad \int e^x dx$$

$$(ii) \quad \int e^{-x} dx$$

$$(iii) \quad \int 4e^{3x} dx$$

$$(iv) \quad \int e^{-x^2 + 4x} dx$$

$$(v) \quad \int \left(2x^3 + 4x^{-3} - 3x + \frac{1}{\sqrt{x}} + \frac{7}{x} \right) dx$$

$$(vi) \quad \int \left(\frac{2x^3 - 3x^2 + 4x - 5}{x^2} \right) dx$$

$$(vii) \quad \int (2 - 3x)(1 + 4x)(7 - 2x) dx$$

$$(viii) \quad \int \left(x^2 + \frac{1}{x^2} \right)^2 dx$$

Solution:

$$(i) \quad \int e^x dx = \frac{e^x}{1} + c = e^x + c$$

$$(ii) \quad \int e^{-x} dx = \frac{e^{-x}}{-1} + c = -e^{-x} + c$$

$$(iii) \quad \int 4e^{3x} dx = \frac{4}{3} e^{3x} + c$$

$$(iv) \quad \int e^{-x^2+4x} dx = \frac{e^{-x^2+4x}}{-2x+4} + c$$

$$(v) \quad \int \left(2x^3 + 4x^{-3} - 3x + \frac{1}{\sqrt{x}} + \frac{7}{x} \right) dx = \frac{2}{4} x^4 + \frac{4}{-2} x^{-2} - \frac{3}{2} x^2 + \frac{x^{1/2}}{1/2} + 7 \ln x + c$$

$$= \frac{1}{2} x^4 - 2x^{-2} - \frac{3}{2} x^2 + 2x^{1/2} + 7 \ln x + c$$

$$(vi) \quad \int \left(\frac{2x^3 - 3x^2 + 4x - 5}{x^2} \right) dx = \int \left(\frac{2x^3}{x^2} - \frac{3x^2}{x^2} + \frac{4x}{x^2} - \frac{5}{x^2} \right) dx$$

$$= \int \left(2x - 3 + \frac{4}{x} - \frac{5}{x^2} \right) dx$$

$$= \frac{2x^2}{2} - 3x + 4 \ln x + \frac{5}{x} + c$$

$$(vii) \quad \int (2-3x)(1+4x)(7-2x) dx = \int (24x^3 - 94x^2 + 31x + 14) dx$$

$$= \frac{24}{4} x^4 - \frac{94}{3} x^3 + \frac{31}{2} x^2 + 14x + c$$

$$= 6x^4 - \frac{94}{3} x^3 + \frac{31}{2} x^2 + 14x + c$$

$$(viii) \quad \int \left(x^2 + \frac{1}{x^2} \right)^2 dx = \int \left(x^4 + \frac{2}{x^4} + \frac{1}{x^4} \right) dx = \frac{x^5}{5} - \frac{2}{3} x^{-3} - \frac{1}{3} x^{-3} + c$$

Exercise

Evaluate the following integrals.

$$(i) \quad \int e^{\frac{1}{4}x} dx$$

$$(ii) \quad \int \left(\frac{1 + \sqrt[3]{x^5}}{\sqrt{x}} \right) dx$$

$$(iii) \quad \int 40x dx$$

$$(iv) \quad \int (9 - 7x + 13x^2) dx$$

$$(v) \quad \int e^{-6x} dx$$

$$(vi) \quad \int \frac{30}{e^{2x}} dx$$

$$(vii) \quad \int \left(\frac{50}{\sqrt[3]{x^2}} \right) dx$$

$$(viii) \quad \int \left(\frac{x^3 + 5\sqrt{x} + 13}{x} \right) dx$$

$$(ix) \quad \int \left(\frac{e^{-3x} - 5e^x + 8}{e^x} \right) dx$$

$$(x) \quad \int \left(x^2 + \frac{1}{x^2} \right)^3 dx$$

Methods of integration

1. Integration by Substitution.

By suitable substitution, the variable x in $\int f(x)dx$ is changed into variable u so that the integrand $f(x)$ is changed into $F(u)$ which is easily integrable.

For example, to evaluate $\int \frac{2x+7}{x^2+7x-13} dx$, we notice that $\frac{d}{dx}(x^2+7x-13) = 2x+7$.

Therefore, we put $u = x^2 + 7x - 13$, then $\frac{du}{dx} = 2x + 7$ or $du = (2x + 7)dx$

$$\therefore \int \frac{2x+7}{x^2+7x-13} dx = \int \frac{1}{u} du = \ln u + c = \ln(x^2 + 7x - 13) + c$$

Note: There is no rule for finding a proper substitute and the best guide in the matter is experience. However, the following suggestions will be useful.

- (i) If the integrand is of the form $f'(ax+b)$, then we put $u = ax+b$ and $\frac{du}{dx} = a$ or

$$du = a dx \text{ or } dx = \frac{1}{a} du.$$

$$\text{Thus } \int f'(ax+b) dx = \int f'(u) \frac{du}{a} = \frac{1}{a} \int f'(u) du = \frac{f(u)}{a} + c = \frac{f(ax+b)}{a} + c$$

- (ii) When the integrand is of the form $x^{n-1} f'(x^n)$.

$$\text{We put } u = x^n \text{ and } \frac{du}{dx} = nx^{n-1} \text{ or } du = nx^{n-1} dx.$$

$$\text{Thus } \int x^{n-1} f'(x^n) dx = \int f'(u) \frac{du}{n} = \frac{1}{n} \int f'(u) du = \frac{1}{n} f(u) + c = \frac{1}{n} f(x^n) + c$$

- (iii) When the integrand is of the form $[f(x)]^n \cdot f'(x)$.

$$\text{We put } u = f(x) \text{ and } du = f'(x) dx$$

$$\text{Thus } \int [f(x)]^n f'(x) = \int u^n du = \frac{u^{n+1}}{n+1} + c = \frac{[f(x)]^{n+1}}{n+1} + c$$

- (iv) When the integrand is of the form $\frac{f'(x)}{f(x)}$.

$$\text{We put } u = f(x) \text{ and } du = f'(x) dx.$$

$$\text{Thus } \int \frac{f'(x)}{f(x)} dx = \int \frac{du}{u} = \ln u + c = \ln f(x) + c$$

- (v) When the integrand is of the form $f'(x)e^{f(x)}$.

$$\text{We put } u = f(x) \text{ and } du = f'(x) dx$$

$$\text{Thus } \int f'(x)e^{f(x)} dx = \int e^u du = e^u + c = e^{f(x)} + c$$

Example

Evaluate the following.

$$(i) \quad \int \frac{2x+3}{x^2+3x+7} dx$$

$$(ii) \quad \int \frac{dx}{\sqrt{x}+x}$$

$$(iii) \quad \int \frac{x^2}{\sqrt[5]{3x^3+7}} dx$$

$$(iv) \quad \int x^4(1+x^5)^{1/3} dx$$

$$(v) \quad \int 2xe^{x^2} dx$$

$$(vi) \quad \int x^2 e^{3x^3} dx$$

$$(vii) \quad \int x^2 \sqrt{x^3+5} dx$$

$$(viii) \quad \int 2x\sqrt{1+x^2} dx$$

$$(ix) \quad \int \frac{1}{\sqrt{2x+1}} dx$$

$$(x) \quad \int e^x \sqrt{1+e^x} dx$$

Solution:

$$(i) \quad \int \frac{2x+3}{x^2+3x+7} dx$$

Let $u = x^2 + 3x + 7$, then $\frac{du}{dx} = 2x + 3$ or $du = (2x + 3)dx$

$$\int \frac{2x+3}{x^2+3x+7} dx = \int \frac{du}{u} = \ln u + c = \ln(x^2 + 3x + 7) + c$$

$$(ii) \quad \int \frac{dx}{\sqrt{x}+x} = \int \frac{dx}{\sqrt{x}(1+\sqrt{x})}$$

Let $u = 1 + \sqrt{x}$, then $\frac{du}{dx} = \frac{1}{2\sqrt{x}}$ or $2du = \frac{dx}{\sqrt{x}}$

$$\int \frac{du}{\sqrt{x}(1+\sqrt{x})} = \int \frac{2du}{u} = 2 \int \frac{1}{u} du = 2 \ln u + c = 2 \ln(1 + \sqrt{x}) + c$$

$$(iii) \quad \int \frac{x^2}{\sqrt[5]{3x^3+7}} dx = \int \frac{x^2}{(3x^3+7)^{1/5}} dx$$

Let $u = 3x^3 + 7$, then $\frac{du}{dx} = 9x^2$ or $du = 9x^2 dx$ or $\frac{du}{9} = x^2 dx$

$$\begin{aligned}
 \int \frac{x^2}{\sqrt[5]{3x^3+7}} dx &= \int \frac{x^2}{(3x^3+7)^{1/5}} dx = \int \frac{du}{9u^{1/5}} = \frac{1}{9} \int u^{-1/5} du = \frac{1}{9} \left[\frac{u^{-1/5+1}}{-1/5+1} \right] + c \\
 &= \frac{1}{9} \left[\frac{u^{4/5}}{4/5} \right] + c \\
 &= \frac{1}{9} \times \frac{5}{4} u^{4/5} \\
 &= \frac{5}{36} (3x^3+7) + c
 \end{aligned}$$

(iv) $\int x^4 (1+x^5)^{1/3} dx$

Let $u = 1+x^5$, then $\frac{du}{dx} = 5x^4$ or $\frac{du}{5} = x^4 dx$

$$\int x^4 (1+x^5)^{1/3} dx = \int u^{1/3} \frac{du}{5} = \frac{1}{5} \int u^{1/3} du = \frac{1}{5} \left[\frac{u^{4/3}}{4/3} \right] + c = \frac{3}{20} u^{4/3} + c = \frac{3}{20} (1+x^5)^{4/3} + c$$

(v) $\int 2xe^{x^2} dx$

Let $u = x^2$, then $\frac{du}{dx} = 2x$ or $du = 2x dx$

$$\int 2xe^{x^2} dx = \int e^u du = \frac{e^u}{1} + c = e^{x^2} + c$$

(vi) $\int x^2 e^{3x^3} dx$

Let $u = 3x^3$, then $\frac{du}{dx} = 9x^2$ or $\frac{du}{9} = x^2 dx$

$$\int x^2 e^{3x^3} dx = \int \frac{1}{9} e^u du = \frac{1}{9} e^u + c = \frac{1}{9} e^{3x^3} + c$$

(vii) $\int x^2 \sqrt{x^3+5} dx = \int x^2 (x^3+5)^{1/2} dx$

Let $u = x^3+5$, then $\frac{du}{dx} = 3x^2$ or $\frac{du}{3} = x^2 dx$

$$\int x^2 \sqrt{x^3+5} dx = \int x^2 (x^3+5)^{1/2} dx = \int \frac{u^{1/2}}{3} du = \frac{1}{3} \left[\frac{u^{3/2}}{3/2} \right] + c = \frac{2}{9} u^{3/2} + c = \frac{2}{9} (x^3+5)^{3/2} + c$$

(viii) $\int 2x \sqrt{1+x^2} dx = \int 2x (1+x^2)^{1/2} dx$

Let $u = 1+x^2$, then $\frac{du}{dx} = 2x$ or $du = 2x dx$

$$\int 2x \sqrt{1+x^2} dx = \int 2x (1+x^2)^{1/2} dx = \int u^{1/2} du = \frac{2}{3} u^{3/2} + c = \frac{2}{3} (1+x^2)^{3/2} + c$$

$$(ix) \quad \int \frac{1}{\sqrt{2x+1}} dx = \int \frac{1}{(2x+1)^{1/2}} dx$$

$$\text{Let } u = 2x+1, \text{ then } \frac{du}{dx} = 2 \text{ or } \frac{du}{2} = dx$$

$$\int \frac{1}{\sqrt{2x+1}} dx = \int \frac{1}{(2x+1)^{1/2}} dx = \int \frac{1}{u^{1/2}} \frac{du}{2} = \frac{1}{2} \int u^{-1/2} du = \frac{1}{2} \left[\frac{u^{1/2}}{1/2} \right] + c = u^{1/2} + c$$

$$= (2x+1)^{1/2} + c$$

$$(x) \quad \int e^x \sqrt{1+e^x} dx = \int e^x (1+e^x)^{1/2} dx$$

$$\text{Let } u = 1+e^x, \text{ then } \frac{du}{dx} = e^x \text{ or } du = e^x dx$$

$$\int e^x \sqrt{1+e^x} dx = \int e^x (1+e^x)^{1/2} dx = \int u^{1/2} du = \frac{2}{3} u^{3/2} + c = \frac{2}{3} (1+e^x)^{3/2} + c$$

Exercise

Evaluate

$$(i) \quad \int \frac{\ln x}{x} dx$$

$$(ii) \quad \int 4x^3 e^{x^4} dx$$

$$(iii) \quad \int (2x+1) e^{(x^2+x)} dx$$

$$(iv) \quad \int x(x^2-3)^4 dx$$

$$(v) \quad \int x \sqrt{1-x^2} dx$$

$$(vi) \quad \int \sqrt{x} \sqrt{1+x^{3/2}} dx$$

$$(vii) \quad \int x^3 (x^4+2)^2 dx$$

$$(viii) \quad \int e^x (1-e^x)^3 dx$$

$$(ix) \quad \int x e^{x^2+1} dx$$

$$(x) \quad \int \frac{x^2}{(x^3+8)^4} dx$$

$$(xi) \quad \int \frac{dx}{e^x+1}$$

$$(xii) \quad \int \frac{1}{x(1+\ln x)} dx$$

$$(xiii) \quad \int (e^x + e^{-x})^2 (e^x - e^{-x}) dx$$

2. Integration by parts

The process of integration of the product of two functions is known as integration by parts.

If u and v are two functions of x , then

$$\int (uv)dx = u \int vdx - \int \left[\frac{du}{dx} \int vdx \right] dx$$

In words, integral of the product of two functions.

= first function $\times$ integral of the second – integral of (differential first $\times$ integral of the second function)

Or

$$\int u dv = uv - \int v du$$

NOTE: While solving problems of integration by parts the following rules may be followed:

- (i) Choose the first and second function in such a way that the derivative of the first function and the integral of the second function can be easily found.
- (ii) In case of integrals of the form $\int f(x).x^n dx$, take x^n as the first function and $f(x)$ as the second function.
- (iii) In case of integrals of the form $\int (\ln x)^n dx$ take 1 as the second function and $(\ln x)^n$ as the first function.
- (iv) Rule of integration by parts may be used repeatedly if required.

Example 1

Evaluate the following.

- (i) $\int x e^x dx$
- (ii) $\int \ln x dx$
- (iii) $\int x e^{3x} dx$
- (iv) $\int (\ln x)^2 dx$

Solution:

(i) $\int x e^x dx$

Let $u = x$, then $\frac{du}{dx} = 1$ or $du = dx$ and

Let $dv = e^x$, then $\int dv = \int e^x$ i.e. $v = e^x$

$$\int u dv = uv - \int v du$$

$$\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + c$$

$$(ii) \int \ln x dx$$

$$\int u dv = uv - \int v du$$

$$\text{Let } u = \ln x, \frac{du}{dx} = \frac{1}{x} \text{ or } du = \frac{dx}{x} \text{ and}$$

let $dv = dx$, on integrating both sides

$$\int dv = \int dx \text{ i.e. } v = x$$

$$\int \ln x dx = x \ln x - \int x \frac{dx}{x}$$

$$= x \ln x - \int dx$$

$$= x \ln x - x + c$$

$$(iii) \int x e^{3x} dx$$

$$\int u dv = uv - \int v du$$

$$\text{Let } u = x, \text{ then } \frac{du}{dx} = 1 \text{ or } du = dx \text{ and}$$

Let $dv = e^{3x} dx$, on integrating we get

$$\text{i.e. } \int dv = \int e^{3x} dx$$

$$v = \frac{e^{3x}}{3}$$

$$\int x e^{3x} dx = \frac{x e^{3x}}{3} - \int \frac{e^{3x}}{3} dx$$

$$= \frac{x e^{3x}}{3} - \frac{1}{3} \int e^{3x} dx$$

$$= \frac{x e^{3x}}{3} - \frac{1}{3} \left(\frac{e^{3x}}{3} \right) + c$$

$$= \frac{x e^{3x}}{3} - \frac{e^{3x}}{9} + c$$

$$(vi) \int (\ln x)^2 dx$$

$$\int u dv = uv - \int v du$$

$$\text{Let } u = (\ln x)^2, \text{ then } \frac{du}{dx} = \frac{2}{x} \ln x \text{ or } du = \frac{2}{x} \ln x dx \text{ and}$$

$$\text{Let } dv = dx, \text{ on integrating we get i.e. } \int dv = \int dx$$

$$v = x$$

$$\begin{aligned}\int (\ln x)^2 dx &= x(\ln x)^2 - \int x \left(\frac{2}{x} \ln x \right) dx \\ &= x(\ln x)^2 - \int 2 \ln x dx \quad \dots\dots\dots(1)\end{aligned}$$

Again applying integration by parts for $\int 2 \ln x dx$

$$\int u dv = uv - \int v du$$

Let $u = \ln x$, then $\frac{du}{dx} = \frac{1}{x}$ or $du = \frac{dx}{x}$ and let $dv = 2 dx$, on integrating both sides

we get $v = 2x$

$$\begin{aligned}\int 2 \ln x dx &= 2x \ln x - \int 2x \frac{dx}{x} \\ &= 2x \ln x - \int 2 dx \\ &= 2x \ln x - 2x + c \quad \dots\dots\dots(2)\end{aligned}$$

The substituting the right hand side of equation (2) into equation (1) gives

$$\begin{aligned}\int (\ln x)^2 dx &= x(\ln x)^2 - \int 2 \ln x dx \\ &= x(\ln x)^2 - \{2x \ln x - 2x\} + c \\ &= x(\ln x)^2 - 2x \ln x + 2x + c\end{aligned}$$

Exercise

Evaluate the following integrals

- | | | |
|-------|----------------------------------|--|
| (i) | $\int x e^{2x} dx$ | Ans. $\frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + c$ |
| (ii) | $\int x^2 e^x dx$ | Ans. $-e^{-x}(x+1) + c$ |
| (iii) | $\int x e^{-x} dx$ | Ans. $e^x(x^2 - 2x + 2) + c$ |
| (iv) | $\int x^3 e^x dx$ | Ans. $(x^3 - 3x^2 + 6x - 6)e^x + c$ |
| (v) | $\int x^2 \ln x dx$ | Ans. $\frac{x^3}{3} \ln x - \frac{x^3}{9} + c$ |
| (vi) | $\int \frac{\ln x}{\sqrt{x}} dx$ | Ans. $2x^{1/2}(\ln x - 2) + c$ |

$$(vii) \quad \int \ln(1+x) dx \quad \text{Ans. } (x+1)\ln(1+x) - x + c$$

3. Integration of Rational Functions

A rational function $\frac{f(x)}{g(x)}$, where $f(x)$ and $g(x)$ are polynomials in x , can be easily integrated by splitting it into partial fractions.

Example

Evaluate the following

$$(i) \quad \int \frac{x}{(x-1)(2x+1)} dx$$

$$(ii) \quad \int \frac{2x+3}{x^2-2x-3} dx$$

$$(iii) \quad \int \frac{1}{x^2-x-6} dx$$

$$(iv) \quad \int \frac{7x^2+3x+1}{x^2+x} dx$$

Solution:

$$(i) \quad \text{Let } \frac{x}{(x-1)(2x+1)} = \frac{A}{x-1} + \frac{B}{2x+1}$$

$$= \frac{A(2x+1) + B(x-1)}{(2x+1)(x-1)}$$

$$\Rightarrow x = A(2x+1) + B(x-1)$$

$$\text{Putting } x=1, \text{ we get } A = \frac{1}{3}$$

$$\text{Putting } x = -\frac{1}{2}, \text{ we get } B = \frac{1}{3}$$

$$\text{Thus } \int \frac{x}{(x-1)(2x+1)} dx = \frac{1}{3} \int \frac{dx}{x-1} + \frac{1}{3} \int \frac{dx}{2x+1}$$

$$= \frac{1}{3} \ln(x-1) + \frac{1}{3} \ln(2x+1) + c$$

$$(ii) \quad \text{Let } \frac{2x+3}{x^2-2x-3} = \frac{2x+3}{(x-3)(x+1)} = \frac{A}{x-3} + \frac{B}{x+1}$$

$$= \frac{A(x+1) + B(x-3)}{(x-3)(x+1)}$$

$$\Rightarrow 2x+3 = A(x+1) + B(x-3)$$

$$\text{Putting } x = -1, \text{ we get } B = \frac{9}{4}$$

Putting $x = 3$, we get $A = \frac{-1}{4}$

$$\begin{aligned}\int \frac{2x+3}{x^2-2x-3} dx &= \frac{9}{4} \int \frac{dx}{x-3} - \frac{1}{4} \int \frac{dx}{x+1} \\ &= \frac{9}{4} \ln(x-3) - \frac{1}{4} \ln(x+1) + c\end{aligned}$$

$$\begin{aligned}\text{(iii) Let } \frac{1}{x^2-x-6} &= \frac{1}{(x-3)(x+2)} = \frac{A}{x-3} + \frac{B}{x+2} \\ &= \frac{A(x+2)+B(x-3)}{(x-3)(x+2)}\end{aligned}$$

$$\Rightarrow 1 = A(x+2) + B(x-3)$$

Putting $x = -2$, we get $B = -\frac{1}{5}$.

Putting $x = 3$, we get $A = \frac{1}{5}$

$$\begin{aligned}\int \frac{1}{x^2-x-6} dx &= \frac{1}{5} \int \frac{dx}{x-3} - \frac{1}{5} \int \frac{dx}{x+2} \\ &= \frac{1}{5} \ln(x-3) - \frac{1}{5} \ln(x+2) + c\end{aligned}$$

(iv) On dividing, we get

$$\frac{7x^2+3x+1}{x^2+x} = 7 + \frac{1-4x}{x^2+x}$$

$$\begin{aligned}\text{Let } \frac{1-4x}{x(x+1)} &= \frac{A}{x} + \frac{B}{x+1} \\ &= \frac{A(x+1)+B(x)}{x(x+1)}\end{aligned}$$

$$\Rightarrow 1-4x = A(x+1) + Bx$$

Putting $x = -1$, we get $B = -5$

Putting $x = 0$, we get $A = 1$

$$\begin{aligned}\int \frac{7x^2+3x+1}{x^2+x} dx &= \int \left(7 + \frac{1-4x}{x(x+1)} \right) dx \\ &= \int 7 dx + \int \frac{1}{x} dx - 5 \int \frac{1}{x+1} dx \\ &= 7x + \ln x - 5 \ln(x+1) + c\end{aligned}$$

Exercise

Evaluate the following integrals

$$(i) \quad \int \frac{(x-1)}{(x-2)(x-3)} dx$$

$$(ii) \quad \int \frac{2x}{(x-1)(x+1)} dx$$

$$(iii) \quad \int \frac{1}{2x^2 + x - 1} dx$$

$$(iv) \quad \int \frac{dt}{2t^2 + 3t + 1}$$

$$(v) \quad \int \frac{x^2 + x - 1}{x^2 + x - 6} dx$$

Applications of Integral Calculus

Integration can be used to determine the area between curves and also to solve business problems.

Finding the area between a curve and the x -axis.

For a function f that is continuous over $[a, b]$, the area between $y = f(x)$ and the x axis from $x = a$ to $x = b$ can be found using definite integrals as follows:

- For $f(x) \geq 0$ over $[a, b]$; $Area = \int_a^b f(x) dx$
- For $f(x) \leq 0$ over $[a, b]$; $Area = \int_a^b -f(x) dx$

Example

Find the area bounded by $f(x) = 6x - x^2$ and $y = 0$ for $1 \leq x \leq 4$.

Solution:

$$Area = \int_1^4 (6x - x^2) dx = \left[3x^2 - \frac{x^3}{3} \right]_1^4 = \left[3(4)^2 - \frac{4^3}{3} \right] - \left[3(1)^2 - \frac{1^3}{3} \right] = 48 - 24 = 24$$

Finding the area between two curves.

If f and g are continuous and $f(x) \geq g(x)$ over the interval $[a, b]$, then the area bounded by $y = f(x)$ and $y = g(x)$ for $a \leq x \leq b$ is given by

$$Area = \int_a^b [f(x) - g(x)] dx$$

Example

Find the area bounded by the graphs of $f(x) = \frac{1}{2}x + 3$, $g(x) = -x^2 + 1$, $x = -2$ and $x = 1$.

Solution

$$\begin{aligned} Area &= \int_a^b [f(x) - g(x)] dx \\ &= \int_{-2}^1 \left[\left(\frac{x}{2} + 3 \right) - (-x^2 + 1) \right] dx \end{aligned}$$

$$\begin{aligned}
&= \int_{-2}^1 \left(x^2 + \frac{x}{2} + 2 \right) dx \\
&= \left[\frac{x^3}{3} + \frac{x^2}{4} + 2x \right]_{-2}^1 \\
&= \left[\frac{1^3}{3} + \frac{1^2}{4} + 2(1) \right] - \left[\frac{(-2)^3}{3} + \frac{(-2)^2}{4} + 2(-2) \right] \\
&= \left(\frac{1}{3} + \frac{1}{4} + 2 \right) - \left(\frac{-8}{3} + \frac{4}{4} - 4 \right) = 8.25
\end{aligned}$$

Definite Integral

Let $f(x)$ and $g(x)$ be two functions of x such that

$$\int f(x) dx = g(x)$$

Then, the definite integral of $f(x)$ over the interval $[a, b]$, denoted by $\int_a^b f(x) dx$ is defined as

$$\int_a^b f(x) dx = [g(x)]_a^b = g(b) - g(a)$$

Where a and b are two real numbers, and are called, respectively, the **lower** and the **upper limits** of the integral.

Example

Evaluate

- (i) $\int_0^1 (x^2 + 1) dx$
- (ii) $\int_0^2 \sqrt{6x + 4} dx$
- (iii) $\int_2^4 \frac{1}{x} dx$
- (iv) $\int_1^2 (e^{3x} + 3x^2) dx$

Solution:

$$\begin{aligned}
 \text{(i)} \quad \int_0^1 (x^2 + 1) dx &= \left[\frac{x^3}{3} + x \right]_0^1 \\
 &= \left[\frac{1^3}{3} + 1 \right] - \left[\frac{0^3}{3} + 0 \right] \\
 &= \frac{1}{3} + 1 - 0 - 0 \\
 &= \frac{4}{3}
 \end{aligned}$$

$$\text{(ii)} \quad \int_0^2 \sqrt{6x+4} dx$$

$$\text{Let } u = 6x + 4, \text{ then } \frac{du}{dx} = 6 \text{ or } dx = \frac{du}{6}$$

$$\begin{aligned}
 \int_0^2 \sqrt{6x+4} dx &= \int_0^2 \sqrt{u} \frac{du}{6} = \frac{1}{6} \int_0^2 u^{\frac{1}{2}} du = \left[\frac{1}{6} \left(\frac{u^{\frac{3}{2}}}{\frac{3}{2}} \right) \right]_0^2 = \left[\frac{1}{6} \times \frac{2}{3} u^{\frac{3}{2}} \right]_0^2 = \left[\frac{1}{9} u^{\frac{3}{2}} \right]_0^2 \\
 &= \left[\frac{1}{9} (6x+4)^{\frac{3}{2}} \right]_0^2 \\
 &= \left[\frac{1}{9} (6 \times 2 + 4)^{\frac{3}{2}} \right] - \left[\frac{1}{9} (6 \times 0 + 4)^{\frac{3}{2}} \right] \\
 &= \frac{1}{9} (16)^{\frac{3}{2}} - \frac{1}{9} (4)^{\frac{3}{2}} \\
 &= \frac{56}{9}
 \end{aligned}$$

$$\text{(iii)} \quad \int_2^4 \frac{1}{x} dx = [\ln x]_2^4 = \ln 4 - \ln 2 = \ln 2^2 - \ln 2 = 2 \ln 2 - \ln 2 = \ln 2$$

$$\text{(iv)} \quad \int_1^2 (e^{3x} + 3x^2) dx = \left[\frac{e^{3x}}{3} + x^3 \right]_1^2 = \left[\frac{e^6}{3} + 8 \right] - \left[\frac{e^3}{3} + 1 \right] = \frac{e^6}{3} + 8 - \frac{e^3}{3} - 1 = \frac{1}{3} (e^6 - e^3) + 7$$

Exercise

- Find the area bounded by the graphs of $f(x) = x^2 - x$, $g(x) = 2x$, for $-2 \leq x \leq 3$.
- Find the area bounded by the graphs of $f(x) = 2x^2$, $g(x) = 4 - 2x$, for $-2 \leq x \leq 2$.
- Find the area bounded by the graph of $f(x) = x^2 - 9$ and the x axis over the indicated intervals

$$\text{(i)} \quad [0, 2]$$

$$\text{(ii)} \quad [2, 4]$$

4. Find the area bounded by $f(x) = x^2 - 1$, $g(x) = \frac{-1}{2}x - 3$, $x = -1$ and $x = 2$.
5. Find the area bounded by the graphs of $f(x) = x^2 - 1$ and $y = 0$ for $-1 \leq x \leq 3$.
6. Integrate each of the following:

$$(i) \int_2^3 \frac{x^2 + 1}{(x^3 + 3x)^2} dx$$

$$(ii) \int_1^3 \frac{8x - 2}{(2x^2 - x + 1)^3} dx$$

$$(iii) \int_0^4 \frac{x + 1}{(x^2 + 2x + 2)^3} dx$$

$$(iv) \int_0^2 (3x^2 + 2x - 1) dx$$

$$(v) \int_0^1 (e^x - e^{-x})^2 dx$$

$$(vi) \int_1^2 x\sqrt{3x - 2} dx$$

$$(vii) \int_1^2 \frac{1}{x^2} e^{-1/x} dx$$

$$(viii) \int_0^1 \frac{3x^3 - 4x^2 + 1}{\sqrt{x}} dx$$

Applications of Integration to cost, revenue and profit functions.

Cost function

The total cost function $C(x)$ is given, then the marginal cost function MC , is the derivative of the total cost function i.e., $MC = \frac{dC}{dx}$. Since integration is the inverse of differentiation, therefore, the total cost function is the integral of the marginal cost function,

$$\text{i.e. } C(x) = \int MC dx + k$$

where k is the arbitrary constant of integration. The value of k can be evaluated if the fixed cost (i.e. the cost when $x = 0$) or the total cost at any arbitrary value is given.

Revenue function

The total revenue function $R(x)$ is given, then the marginal revenue function MR , is the derivative of the total revenue function i.e., $MR = \frac{dR}{dx}$. Since integration is the inverse of differentiation, therefore, the total revenue function is the integral of the marginal revenue function,

$$\text{i.e. } R(x) = \int MR dx + k$$

where k is the arbitrary constant of integration, which can be evaluated from the fact that the total revenue $R(x)$ is zero when the output is zero.

Also we know that $R(x) = px$, where p is the price $\Rightarrow p = \frac{R(x)}{x}$, which is the demand function.

Example

1. The marginal cost function of manufacturing x shoes is $6 + 10x - 6x^2$. The total cost of producing a pair of shoes is £12. Find the total and average cost function.

Solution:

We know that

$$MC = \frac{d}{dx}(TC)$$

$$\text{Thus } TC = \int MC dx + k$$

$$= \int (6 + 10x - 6x^2) dx + k$$

$$\therefore C(x) = 6x + \frac{10}{2}x^2 - \frac{6}{3}x^3 + k$$

Where k is the constant of integration.

Also we are given when $x = 2$, then $C(x) = 12$

$$\therefore 12 = 6 \times 2 + 10 \times \frac{4}{2} - \frac{6}{3} \times 2^3 + k$$

$$12 = 12 + 20 - 16 + k$$

$$k = -4$$

Hence the total cost function $C(x) = 6x + 5x^2 - 2x^3 - 4$

AC, by definition, is $\frac{C(x)}{x} = 6 + 5x - 2x^2 - \frac{4}{x}$

2. If the marginal cost function is given by $MC = 2(2x + 9)^{-1/2}$ and the fixed cost is £4, find the average cost for 8 units of output.

Solution:

The marginal cost is

$$MC = 2(2x + 9)^{-1/2}$$

$$C(x) = \int 2(2x + 9)^{-1/2} dx$$

$$\text{Let } u = 2x + 9, \text{ then } \frac{du}{dx} = 2 \text{ or } dx = \frac{du}{2}$$

$$C(x) = \int 2(2x + 9)^{-1/2} dx = \int 2u^{-1/2} \frac{du}{2} = \int u^{-1/2} du = \frac{u^{1/2}}{1/2} + k = 2u^{1/2} + k$$

$$\therefore C(x) = 2(2x + 9)^{1/2} + k$$

Where k is the constant of integration.

$$\text{Since } C(0) = 4, \text{ we have } 4 = 2(2 \times 0 + 9)^{1/2} + k$$

$$4 = 2(9)^{1/2} + k$$

$$4 = 2 \times 3 + k$$

$$k = -2$$

$$\text{Hence } C(x) = 2\sqrt{2x + 9} - 2$$

The average cost function is given by

$$AC = \frac{C(x)}{x} = \frac{2\sqrt{2x + 9}}{x} - \frac{2}{x}$$

$$\text{For } x = 8, AC = \frac{2\sqrt{2(8)+9}}{8} - \frac{2}{8} = 1$$

3. A manufacturer's marginal cost function is $MC = 0.003x^2 - 0.6x + 40$, where x is the number of units of a product. If x increases from 100 to 200 units, find the total increase in cost.

Solution:

$$MC = 0.003x^2 - 0.6x + 40$$

The total increase in cost when x increases is

$$\begin{aligned} &= C(200) - C(100) \\ &= \int_{100}^{200} MC dx \\ &= \int_{100}^{200} (0.003x^2 - 0.6x + 40) dx \\ &= \left[0.003 \frac{x^3}{3} - 0.6 \frac{x^2}{2} + 40x \right]_{100}^{200} \\ &= [8000 - 12000 + 8000] - [1000 - 3000 + 4000] \\ &= 2000 \end{aligned}$$

4. Marginal cost of production in dollars is $3 + \frac{x}{3000} + e^{-0.03x}$, where x denotes the number of units. Find the cost of producing 100 units. It is given that $e^{-3} = 0.05$ (approximately).

Solution:

The marginal cost function is

$$MC = 3 + \frac{x}{3000} + e^{-0.03x}$$

Total cost of producing 100 units is

$$\begin{aligned} \int_0^{100} MC dx &= \int_0^{100} \left(3 + \frac{x}{3000} + e^{-0.03x} \right) dx \\ &= \left[3x + \frac{x^2}{6000} - \frac{e^{-0.03x}}{0.03} \right]_0^{100} \\ &= 300 + \frac{10000}{6000} - \frac{e^{-3}}{0.03} - \left(\frac{-e^0}{0.03} \right) \\ &= 300 + \frac{5}{3} - \frac{5}{3} + \frac{1}{0.03} \quad [\text{As } e^{-3} = 0.05] \\ &= 300 + \frac{100}{3} = 333.33 \end{aligned}$$

5. The function describing the marginal cost of producing a product is $MC = x + 100$ where x equals the number of units produced. It is also known that total cost equals Ksh. 40,000 When $x = 100$. Determine the total cost function.

Solution:

To determine the total cost, we must find the antiderivative of the marginal cost function,

$$\begin{aligned} \text{or } C(x) &= \int MC dx \\ &= \int (x + 100) dx \\ &= \frac{x^2}{2} + 100x + c \end{aligned}$$

$$\therefore C(x) = \frac{x^2}{2} + 100x + c$$

Given that $C(100) = 40,000$, we can solve for the value of c , which happens to represent the fixed cost.

$$\begin{aligned} 40,000 &= \frac{100^2}{2} + 100(100) + c \\ 40,000 &= 5,000 + 10,000 + c \\ c &= 25,000 \end{aligned}$$

The specific function representing the total of producing the product is

$$\therefore C(x) = \frac{x^2}{2} + 100x + 25,000$$

6. A state organization is conducting its annual fund-raising campaign for its summer camp program for the disadvantaged. Campaign expenditures will be incurred at a rate of £ 10,000 per day. From past experience it is known that contribution will be high during the early stages of the campaign and will tend to fall off as the campaign continues. The function describing the rate at which contribution are received is $C(t) = -100t^2 + 20,000$ where t represents the day of the campaign, and $C(t)$ equals the rate at which contributions are received, measured in shillings per day. The organization wishes to maximize the net proceeds from the campaign.

- (i) Determine how long the campaign should be conducted in order to maximize net proceeds.
- (ii) What are the total campaign expenditures expected to equal?
- (iii) What are the total contributions expected to equal?
- (iv) What are the net proceeds (total contribution less expenditures) expected to equal?

Solution:

- (i) The function which describes the rate at which expenditures $e(t)$ are incurred is

$$e(t) = 10,000$$

$$C(t) = e(t)$$

$$-100t^2 + 20,000 = 10,000$$

$$t^2 = 100$$

$$t = 10 \text{ days (negative root is meaningless)}$$

- (ii) Total campaign expenditures are represented by

$$E = e(t)$$

$$= \{10,000 \text{ per day}\} \times \{10 \text{ days}\}$$

$$= 100,000$$

- (iii) Total contribution during the 10 days are represented by

$$C = \int_0^{10} (-100t^2 + 20,000) dt$$

$$= -100 \frac{t^3}{3} + 20,000t \Big|_0^{10}$$

$$= -100 \frac{(10)^3}{3} + 20,000(10)$$

$$= 166,666.67$$

- (iv) $\text{Netproceeds} = \text{Costs} - \text{Expenditures}$

$$= 166,666.67 - 100,000$$

$$= 66,666.67$$

7. Gatheru and Kabiru Public Accountants have recently started to give advice to their clients. Acting as consultants, they have estimated the demand curve of a client's firm to be $AR = 200 - 8Q$ where AR is the average revenue in shillings and Q is the output in units. Investigators have shown that the firm's cost profile shows that marginal cost MC is given by $MC = Q^2 - 28Q + 211$ (in millions of shillings). Investigations have shown that the firm's cost when not producing output is $Ksh.10$ million.

Required:

- (i) The total cost function
- (ii) The total revenue function
- (iii) The level of output that maximize profit

Solution:

- (i) The total cost function

$$TC = \int MC dq = \int (q^2 - 28q + 211) dq = \frac{1}{3}q^3 - 14q^2 + 211q + c$$

$$\therefore C(x) = \frac{1}{3}q^3 - 14q^2 + 211q + c$$

$$\text{Given that } C(0) = 10, \text{ then } C(0) = \frac{1}{3}(0)^3 - 14(0)^2 + 211(0) + c$$

$$10 = c$$

$$\therefore C(x) = \frac{1}{3}q^3 - 14q^2 + 211q + 10$$

(ii) The total revenue function

$$TR = AR \times Q = (200 - 8q)q = 200q - 8q^2$$

(iii) The level of output that maximizes profit

$$\text{Profit} = TR - TC = 200q - 8q^2 - \left(\frac{1}{3}q^3 - 14q^2 + 211q + 10\right)$$

$$= 200q - 8q^2 - \frac{1}{3}q^3 + 14q^2 - 211q - 10$$

$$P = -\frac{1}{3}q^3 + 6q^2 - 11q + 10$$

To maximize profit, $\frac{dP}{dq} = 0$ and $\frac{d^2P}{dq^2} < 0$

$$\frac{dP}{dq} = -q^2 + 12q - 11 = 0$$

$$\therefore q = \frac{-12 \pm \sqrt{12^2 - (4 \times -1 \times -11)}}{-2} = \frac{-12 \pm 10}{-2}$$

$$q = \frac{-12 + 10}{-2} = 1 \text{ or } q = \frac{-12 - 10}{-2} = 11$$

$$\frac{d^2P}{dq^2} = -2q + 12$$

When $q = 1$, $\frac{d^2P}{dq^2} = -2(1) + 12 = 10 > 0$ hence minimum.

When $q = 11$, $\frac{d^2P}{dq^2} = -2(11) + 12 = -10 < 0$ hence maximum.

$\therefore$ the level of output that will maximize profit is 11 units.

8. If the marginal revenue is given by $15 - 2x - x^2$, find the total revenue and demand function. Find also the maximum revenue.

Solution:

We know that $MR = \frac{d}{dx}(TR)$

$$\Rightarrow TR = \int MR dx + k$$

$$\text{Thus, } TR = \int (15 - 2x - x^2) dx + k$$

$$R(x) = 15x - x^2 - \frac{x^3}{3} + k$$

At $x = 0$, $R(0) = 0$ and thus $k = 0$.

$$\text{Hence } R(x) = 15x - x^2 - \frac{x^3}{3}$$

If p is the demand function, then $TR = px$

$$\Rightarrow p = \frac{TR}{x} = 15 - x - \frac{x^2}{3}$$

Again, for maximum revenue

$$\frac{d}{dx}(TR) = 0$$

$$\Rightarrow 15 - 2x - x^2 = 0$$

$$x = -5 \text{ or } 3$$

Since $x = -5$ is not possible, we take $x = 3$.

Also then, maximum revenue

$$= 15 \times 3 - 9 - \frac{27}{3} = 27$$

9. The marginal revenue for a company's product is $MR = 50,000 - x$, where x equals the number of units produced and sold. If the total revenue equals 0 when no unit are sold, determine the total revenue function for the producer.

Solution:

Because the marginal revenue function is the derivative of the total revenue function, the total revenue function is the antiderivative of the marginal revenue function

$$\begin{aligned} TR &= \int MR dx + k \\ &= \int (50,000 - x) dx + k \end{aligned}$$

$$R(x) = 50,000x - \frac{x^2}{2} + k$$

Since we are told that $R(0) = 0$, substitution of $x = 0$ and $R = 0$ into the above equation yields

$$\begin{aligned} 0 &= 50,000(0) - \frac{0^2}{2} + k \\ k &= 0 \end{aligned}$$

Thus the total revenue function for the company's product is $R(x) = 50,000x - \frac{x^2}{2}$.

10. An automobile manufacturer estimates that the annual rate of expenditure $r(t)$ for maintenance on one of its model is represented by the function $r(t) = 100 + 10t^2$.
- (i) What is the expected maintenance expenditures during the automobile first 5 years?
 - (ii) Of the above expenditures, what is expected to be incurred during the fifth year?

Solution:

$$(i) \quad \int_0^5 (100 + 10t^2) dt = \left[100t + \frac{10t^3}{3} \right]_0^5 = 100(5) + \frac{10(5)^3}{3} = 500 + 416.67 = 916.67$$

(ii) Of these expenditures, those expected to be incurred during the fifth year are estimated as

$$\begin{aligned} \int_4^5 (100 + 10t^2) dt &= \left[100t + \frac{10t^3}{3} \right]_4^5 \\ &= \left[100(5) + \frac{10(5)^3}{3} \right] - \left[100(4) + \frac{10(4)^3}{3} \right] \\ &= 916.67 - [400 + 213.33] \\ &= 303.34 \end{aligned}$$

11. If the marginal revenue function for a commodity is $MR = 9 - 4x^2$, find the demand function.

Solution:

We have $MR = 9 - 4x^2$

Thus $R(x) = \int MR dx + k$

$$\int (9 - 4x^2) dx + k$$

$$R(x) = 9x - \frac{4}{3}x^3 + k$$

Since $R(x) = 0$, when $x = 0$

$$\therefore k = 0$$

$$\text{Thus } R(x) = 9x - \frac{4}{3}x^3$$

If p is the price $R(x) = px$ and so the demand function is $p = \frac{R(x)}{x} = 9 - \frac{4}{3}x^2$

12. The marginal revenue (in thousands of dollars) function for a particular commodity is $5 + 3e^{-0.03x}$ where x denotes the number of units sold. Determine the total revenue from the sale of 100 units. Given $e^{-3} = 0.05$ approximately.

Solution:

We have $MR = 5 + 3e^{-0.03x}$

$$R(x) = \int MR dx + k$$

$$= \int_0^{100} (5 + 3e^{-0.03x}) dx$$

$$\begin{aligned}
&= \left[5x + 3 \frac{e^{-0.03x}}{(-0.03)} \right]_0^{100} \\
&= \left[5 \times 100 - \frac{3}{0.03} e^{-3} \right] - \left[5 \times 0 - \frac{3}{0.03} e^0 \right] \\
&= 500 - 100e^{-3} + 100e^0 \\
&= 500 - 100(0.05) + 100 \\
&= 595 \\
\therefore &59500
\end{aligned}$$

13. A manufacturer's marginal revenue function $MR = 275 - x - 0.3x^2$. Find the increase in the manufacturer's total revenue if production is increased from 10 to 20 units.

Solution:

The marginal revenue is

$$MR = 275 - x - 0.3x^2 \text{ i.e. } \frac{dR}{dx} = 275 - x - 0.3x^2$$

We are required to find total increase in revenue when x increases from 10 to 20 units, i.e., we have to find $R(20) - R(10)$

$$\begin{aligned}
\text{Now } R(20) - R(10) &= \int_{10}^{20} MR dx \\
&= \int_{10}^{20} (275 - x - 0.3x^2) dx \\
&= \left[275x - \frac{x^2}{2} - 0.3 \frac{x^3}{3} \right]_{10}^{20} \\
&= \left[275 \times 20 - \frac{20^2}{2} - 0.3 \frac{(20)^3}{3} \right] - \left[275 \times 10 - \frac{10^2}{2} - 0.3 \frac{(10)^2}{3} \right] \\
&= 1900
\end{aligned}$$

Exercise

1. If the marginal revenue is given by $MR = 7 - 4x - x^2$, find the total revenue and demand function.
2. If the marginal revenue function for output x is given by $MR = \frac{6}{(x+2)^2} + 5$, find the total revenue function and the demand equation.
3. The marginal revenue function of a monopolist is given as $R'(x) = 50 - 0.0002x^2$, where R' denotes marginal revenue and x denotes the quantity produced and sold. It is known

that total revenue is zero when $x = 0$. Find the market demand function for the commodity.

4. A company is specializing in mail order sales approach is beginning a promotional campaign. Advertising expenditures will cost the firm £ 5,959 per day. Marketing specialists estimates that the rate at which profit (exclusive of advertising costs) will be generated from the promotion campaign decreases over the length of the campaign. Specifically, the rate $r(t)$ for this campaign is estimated by the function

$$r(t) = -50t^2 + 10,000$$

Where t represents the day of the campaign and $r(t)$ is measured in dollars per day. In order to maximize net profit, the firm should conduct the campaign as long as $r(t)$ exceeds the daily advertising cost.

- (i) Graph the function $r(t)$ and the function $C(t) = 5,959$ which describes the rate at which advertising expenses are incurred.
 - (ii) How long should the campaign be conducted?
 - (iii) What are the total advertising expenditures expected to equal during the campaign?
 - (iv) What net profit will be expected?
5. A manufacturer's marginal revenue function is: $\frac{dR}{dx} = \frac{2000}{\sqrt{400x}}$. If x is in dollars, find the change in the manufacturer's total revenue if production is increased from 400 to 900 units.
6. The marginal cost is given by $MC = 150 + e^{5x}$, where x is the number of units, find the average cost function.
7. The marginal cost function of a product is given by $10 - 0.01x + 0.0009x^2$. Find the total cost function and the average cost function if the cost of producing 10 units is 105 dollars.
8. The marginal cost function is $MC = 2 + 5e^x$.
- (i) Find c , if $C(0) = 100$.
 - (ii) Find AC .
 - (iii) Evaluate c , MC and AC for $x = 60$.
9. If the marginal cost function is given by $MC = 3(3x + 4)^{1/2}$ and fixed cost is 2 dollars, find the average cost for 7 units of output.
10. The marginal cost of production is found to be $MC = 1000 - 20x + x^2$, where x is the number of units produced. The fixed cost of production is 9000 dollars. Find the cost function.
11. If the marginal revenue function for output x is given by $MR = \frac{4}{(2x + 3)^2} - 1$. Show that the average revenue function is equal to $p = \frac{4}{6x + 9} - 1$.